7.1 Classification of Ordinary Differential Equations

Name _____ Period ____ Date _____

AP Calculus 7.1 · Student sheet

I can…

- state the order of a differential equation and whether it is linear.
- verify that a function is a solution by substituting.
- find a general solution and a particular solution with an initial condition.

Vocabulary · fill in the missing word

Differential equation — an equation that contains a _____ of an unknown function

Order — the order of the highest _____ in the equation

General solution — the family of all solutions, written with a _____ C

Particular solution — the one solution that _____ an initial condition

Part 1 · Guess a function

Find a function that works, then check.

Equation	A function y that works	Check
$y' = y$		
$y' = 2x$		
$y'' = -y$		

Part 2 · Notes and examples (fill in each note, then work the example)

Order: the order of the highest _____ in the equation.

Linear: y and its derivatives appear only to the _____ power, multiplied by functions of x.

Example 1

State the order and linearity of $\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 - y = 0$.

Verify a solution: _____ the function and its derivatives into the equation.

Example 2

For which k is $y = e^{kx}$ a solution of $y'' - y' - 6y = 0$?

Particular solution: use the initial condition to find _____.

Example 3

Solve $\frac{dy}{dx} = e^x + 4\sin(2x)$ with $y(0) = 4$.

Example 4

Show that $y = 3\sin(2x)$ is a solution of $y'' + 4y = 0$.

Part 3 · Partner practice

Partner 1

State the order and linearity of $y'' + 3y' - 4y = \sin x$.

Partner 2

(AP® Practice) The general solution of $\frac{dy}{dx} = 2x + \cos x$ is

(A) $y = x^2 + \sin x + C$ (B) $y = x^2 - \sin x + C$

(C) $y = 2 - \sin x + C$ (D) $y = x^2 + \sin x$